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Comptia N+ Assignment

[N+ - Network security, Maintenance and Troubleshooting procedures]

Troubleshooting Tools

Task:

❖ **1. Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.**

➢ **Aim**

- To use the ipconfig command in Windows Command Prompt to identify the computer's IPv4 address and Default Gateway.

➢ **Introduction**

➢ **IPConfig (Internet Protocol Configuration)** is a command-line network troubleshooting tool available in Windows operating systems.

➢ It displays the network configuration details of a computer, including:

- IPv4 Address
- Subnet Mask
- Default Gateway
- Network adapter information

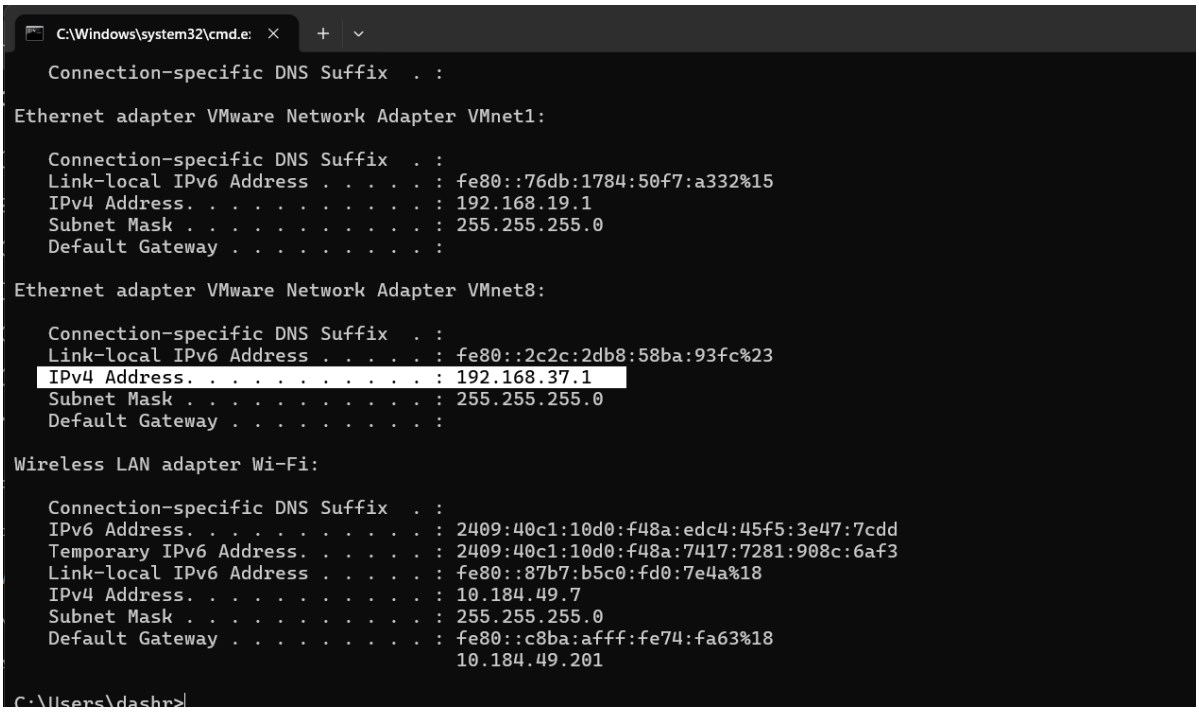
- The ipconfig command helps users check their computer's IP configuration and troubleshoot basic network connectivity problems.

- **Requirements**
  1) A Windows computer or laptop.
  2) An active network connection (Wi-Fi or Ethernet).
  3) Windows Command Prompt.

- **Procedure**
- Follow these steps to find the IPv4 address and Default Gateway:
  1) Turn on the computer and connect it to a Wi-Fi network or Ethernet connection.
  2) Press Windows + R on the keyboard.
  3) Type cmd in the Run window.
  4) Press Enter to open Command Prompt.
  5) Type the following command:
     ipconfig
  6) Press Enter to execute the command.
  7) Find the active network adapter, such as **Wireless LAN adapter Wi-Fi** or **Ethernet adapter**.
  8) Identify the following details:
     - IPv4 Address
     - Subnet Mask
     - Default Gateway
  9) Record the IPv4 Address and Default Gateway shown on your computer.

- **Observation**
- After executing the ipconfig command, Windows displays the network configuration details.

No.	Network Parameter	Description
1	IPv4 Address	Identifies the computer on the IPv4 network.
2	Subnet Mask	Identifies the network portion and host portion of the IP address.
3	Default Gateway	The IP address of the router or gateway used to reach other networks.



- **My Computer's Network Details:**
- IPv4 Address: 192.168.37.1
- Subnet Mask: 255.255.255.0

- **Result**
- The ipconfig command was successfully executed in Windows Command Prompt. The computer's IPv4 address, Subnet Mask, and Default Gateway were identified.

❖ **2. Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.**

- **Aim**
  - To use the ping command in Windows Command Prompt to check network connectivity between a computer and www.google.com and record the average response time.

- **Introduction**
  - **Ping (Packet Internet Groper)** is a network troubleshooting tool used to check whether a destination device or server is reachable over an IP network.
  - It sends ICMP Echo Request packets to the destination and waits for ICMP Echo Reply packets.

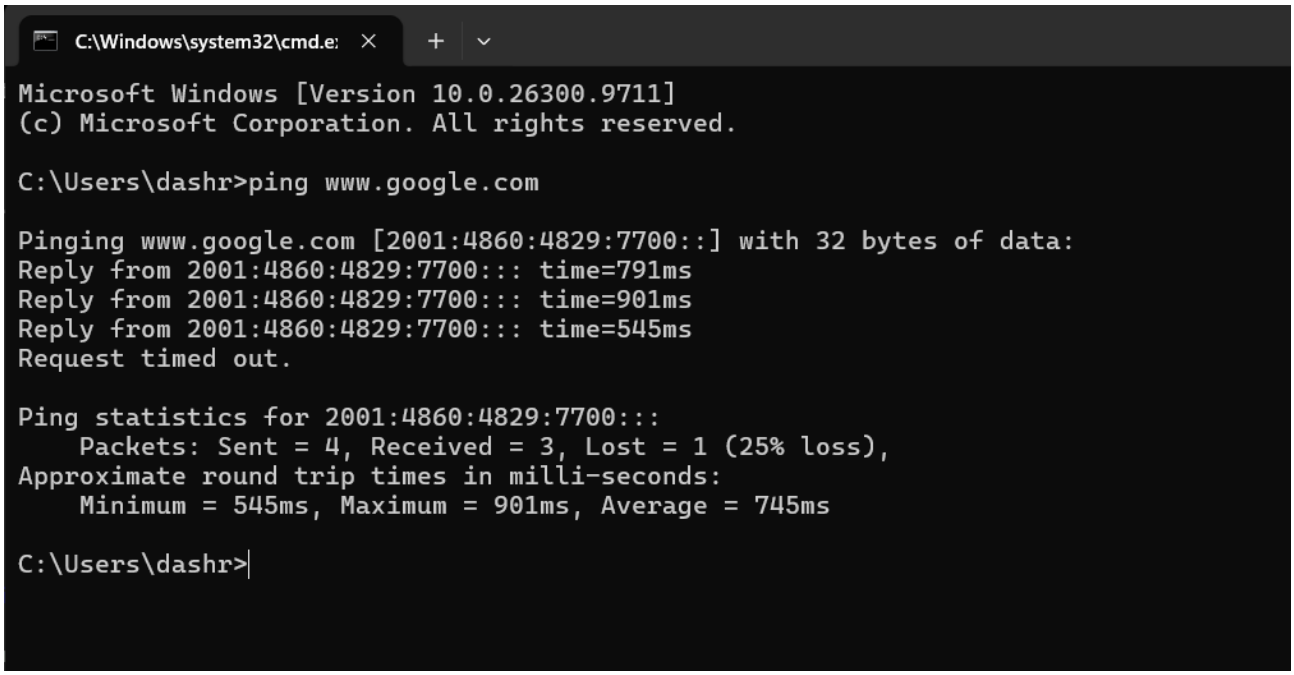
- The ping command helps us check:
  - Network connectivity.
  - Response time.
  - Packet loss.
  - Whether the destination is reachable.

- **Requirements**
  1. A computer or laptop.
  2. An active Internet connection.
  3. Windows Command Prompt.

- **Procedure**
- Follow these steps to check connectivity with Google:
  1. Connect your computer to the Internet using Wi-Fi or Ethernet.
  2. Press Windows + R on the keyboard.
  3. Type cmd and press Enter.
  4. In Command Prompt, type the following command:
     ping www.google.com
  5. Press Enter to execute the command.
  6. Windows sends four ICMP Echo Request packets by default.
  7. Observe the replies received from Google's server.
  8. At the end of the output, check the statistics and note the **Average response time**.

- **Observation**
- After executing the ping command, the output displays the following information:

No.	Parameter	Description
1	Destination	The website being tested (www.google.com).
2	Packets Sent	Total number of packets sent to the destination.
3	Packets Received	Number of reply packets received.
4	Packet Loss	Percentage of packets that did not receive a reply.
5	Average Response Time	Average time taken to receive replies, measured in milliseconds (ms).



- **My Ping Test Results:**
- Destination: www.google.com
- Packets Sent: 4
- Packets Received: 3
- Packet Loss: 1
- Average Response Time: 745ms

- **Result**
- The ping command was successfully executed to test connectivity with www.google.com. The packet transmission results, packet loss, and average response time were observed.

❖ 3. Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

➢ Aim

- To use the tracert command to trace the network route to www.flipkart.com and identify the number of hops required to reach the destination.

➢ Introduction

- **Tracert (Traceroute)** is a network troubleshooting command used to identify the path taken by data packets from a source computer to a destination server.
- It displays the intermediate routers, also known as **hops**, through which packets travel to reach the destination.
- In this practical, the tracert command is used to trace the route from our computer to the Flipkart website and determine the number of hops.

➢ Requirements

- A computer or laptop
- Internet connection
- Command Prompt (Windows) or Terminal (Linux/Mac)

➢ Procedure

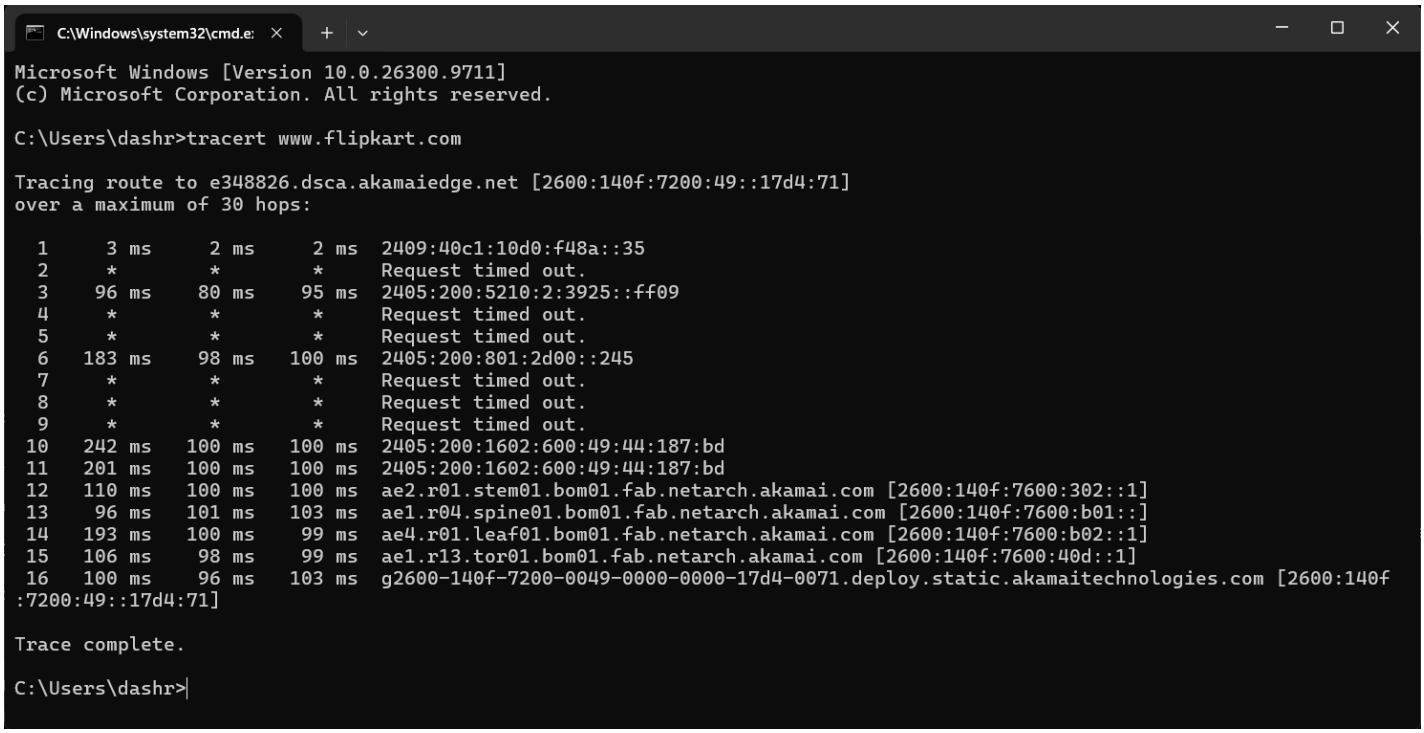
1. Connect the computer to the Internet.
2. Press Windows + R to open the Run window.
3. Type cmd and press Enter to open Command Prompt.

4. Enter the following command:
   tracert www.flipkart.com
5. Press Enter and wait for the command to complete.
6. Observe the list of intermediate routers displayed on the screen.
7. Identify the final destination and count the numbered hops shown in the output.
8. Record the total number of hops required to reach the destination.

- **For Linux or macOS:** Use the command traceroute www.flipkart.com in Terminal. If the command is unavailable, the traceroute utility may need to be installed.

- **Observation**
- The tracert command displays the route taken by packets from the computer to the Flipkart website. Each numbered row represents a hop, usually an intermediate router along the route.

Particulars	**Observation**
Destination Website	www.flipkart.com
Command Used	tracert www.flipkart.com
Total Number of Hops	16
Destination Reached	Yes



- **Result**
- The tracert command was successfully used to trace the network route to www.flipkart.com. The number of hops observed in the command output was **16.**

❖ 4. Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

➢ Aim

- To use the nslookup command to find the IP address of www.spotify.com and understand how this tool helps verify DNS resolution.

➢ Introduction

- **NSLookup (Name Server Lookup)** is a network troubleshooting command used to obtain DNS (Domain Name System) information about a domain name.
- It helps find the IP address associated with a domain name by querying a DNS server. DNS makes it possible for users to access websites using easy-to-remember domain names instead of numerical IP addresses.
- In this practical, the nslookup command is used to find the IP address of www.spotify.com.

➢ Requirements

- A computer or laptop
- Internet connection
- Command Prompt (Windows) or Terminal (Linux/Mac)

➢ Procedure

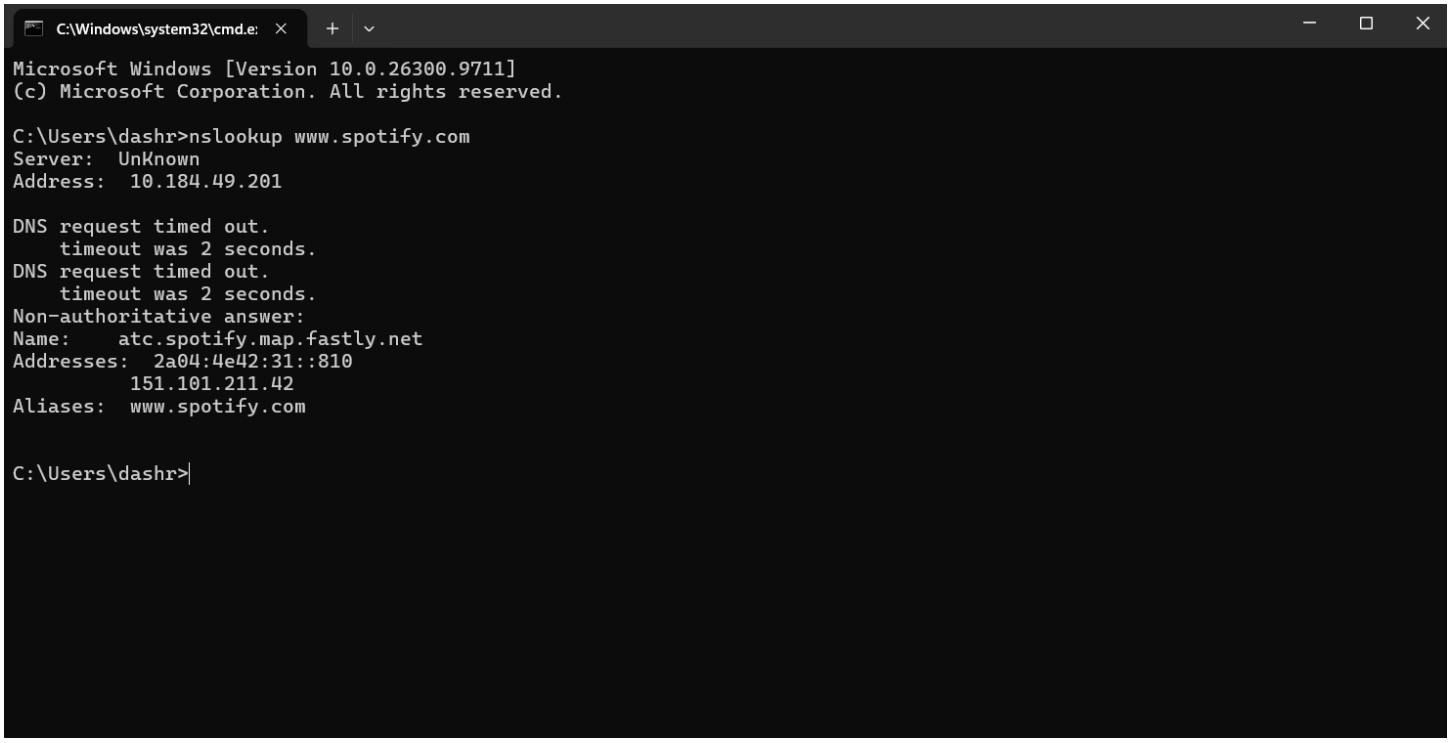
1. Connect the computer to the Internet.
2. Press Windows + R to open the Run window.
3. Type cmd and press Enter.
4. Enter the following command:

nslookup www.spotify.com

5. Press Enter to execute the command.
6. Observe the DNS server information and the IP address displayed in the output.
7. Record the IPv4 address and IPv6 address, if available.

- **For Linux or macOS:** Open Terminal and execute the same command.

- **Observation**
- After executing the command, the DNS server returns the IP address information associated with www.spotify.com.



Particulars	Observation
Domain Name	www.spotify.com
Command Used	nslookup www.spotify.com
DNS Server	Unknown (10.184.49.201)
IPv4 Address	151.101.211.42
IPv6 Address (if available)	2a04:4e42:31::810

- **What Does NSLookup Verify?**
- NSLookup helps verify whether a domain name is successfully resolved into its corresponding IP address through DNS.

- **Result**
- The nslookup command was used to query the DNS information of www.spotify.com and identify its associated IP address.

❖ **5. A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check. Think about checking both network connectivity and DNS resolution.**

➢ **Aim**

- To identify the possible reason why a laptop cannot access the Zomato website by using three network troubleshooting commands.

➢ **Introduction**

- Network troubleshooting commands help identify problems related to network configuration, Internet connectivity, DNS resolution, and routing.
- If a user cannot access the Zomato website, commands such as ipconfig, nslookup, and ping can be used in a systematic sequence to diagnose the issue.

➢ **Troubleshooting Commands**

➢ The following three commands can be used to diagnose the problem:

➢ **1. IPConfig**

➢ **Command:**

- ipconfig

➢ **Purpose:** This command displays the computer's network configuration, including its IPv4 address, subnet mask, and default gateway.

➢ **What it helps check:** It helps verify whether the laptop has a valid IP address and is properly connected to the local network.

- **2. NSLookup**
- **Command:**
  - nslookup www.zomato.com
- **Purpose:** This command queries the DNS server to find the IP address associated with the Zomato domain name.
- **What it helps check:** It helps verify whether DNS is successfully resolving www.zomato.com into an IP address.
- **3. Ping**
- **Command:**
  - ping www.zomato.com
- **Purpose:** This command sends ICMP echo requests to the destination to check connectivity and measure response time.
- **What it helps check:** It helps determine whether the destination responds over the network and whether packet loss or high response times are occurring.

- **Sequence of Troubleshooting**

1) First, run ipconfig to check the laptop's IP address and default gateway.
2) Next, run nslookup www.zomato.com to check whether the Zomato domain resolves to an IP address.
3) Finally, run ping www.zomato.com to check whether the destination responds to network requests.

- **Result**
- The three troubleshooting commands—IPConfig, NSLookup, and Ping—can be used to check local network configuration, DNS resolution, and network connectivity when a user cannot access the Zomato website.